AI2ORBIT PROJECT · REPORT ENGINE

UK Mobile Data Market

A comprehensive analysis of pricing, handset ex-factory margins, and marketing economics — with a machine-readable sample feed

Prepared for Sami Leino · 2026-08-01 · 18 tariffs · 11 handsets · 13 market metrics

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Executive Summary

The UK mobile-data market is built on four physical networks — EE, O2, Vodafone and Three — resold through a dense layer of MVNOs. The headline finding of this report is the size of the **per-GB spread**: the cheapest effective data (iD Mobile 120GB, ≈£0.10/GB, on Three) is roughly **19× cheaper per gigabyte** than an own-brand small bundle (O2 3GB, ≈£1.93/GB). Large bundles on reseller brands crush the per-GB rate; network-owned own-brand plans carry a 3–15× premium for brand, support and roaming certainty.

On devices, flagship handsets sell at roughly **2.3–2.8× their bill-of-materials** (ex-VAT), a gross margin of 56–64% of trade value. That margin is the pool from which assembly, software, IP/royalties, logistics, warranty and marketing are paid — it is *not* net profit. The multiple compresses toward the budget tiers (~1.6×). Real ex-factory prices are confidential, so the ex-factory column here is a clearly-flagged model (BOM × 1.30).

On marketing, UK carriers spend an estimated **15–20% of revenue** on customer acquisition and retention; digital ad spend was ~$420M in 2023, with challenger brands (Smarty +90%, Three +60% YoY) spending hardest. eSIM is reshaping unit economics — eliminating plastic-SIM logistics is estimated to cut acquisition cost ~30% — and online now drives ~35% of acquisition.

Every figure in this report is tagged measured or modelled. Measured figures are reported (see sources); modelled figures are computed from the stated assumptions below and are never presented as quotes.

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Methodology & Honesty Labels

Basis of every figure

- **measured** — a public or reported figure; the source is named in the table or the Sources section.
- **modelled** — computed here from a stated assumption; not a quoted number. Modelled rows and cells are shown in **orange**.

Modelling constants

Constant	Value	Used for
FX_USD_GBP	0.79	USD→GBP on handset BOM / ex-factory
EXFACTORY_MULT	1.30	ex-factory ≈ BOM × 1.30 (assembly, test, maker margin)
VAT_RATE	0.20	stripping UK VAT from RRP to a trade value

What is measured vs modelled here

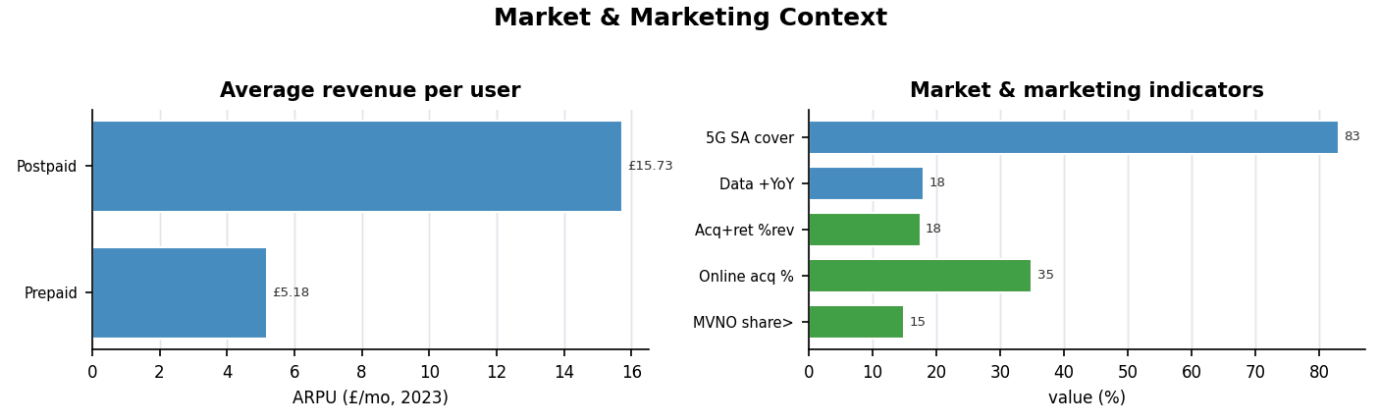
- SIM-only prices — measured list prices, mid-2026 (two tiers interpolated and flagged modelled).
- Handset BOM — measured teardown estimates (analyst-dependent); ex-factory modelled.
- ARPU, coverage, ad spend, acquisition economics — measured, reported figures.
- Cost-per-acquisition (£) — modelled placeholder pending the client's own figure.

> Because a real market has hundreds of tariffs and confidential device costs, a genuinely exhaustive report mixes measured anchors with clearly-flagged models. This report never fills a page by inventing "facts" — length comes from real data, operator/handset profiles and derived analysis.

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Market Overview

The UK is a mature, four-MNO market with a large and growing MVNO layer. Virgin Media O2 alone reported 45.7M total connections at end-2024; 5G standalone now reaches ~83% of the population, and mobile data usage is still growing ~18% year-on-year.



Revenue per user remains modest — postpaid ARPU ~£15.73/month, prepaid ~£5.18 (2023) — which is what makes the per-GB economics and acquisition costs on the following pages so decisive for operator profitability.

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SIM-Only Pricing — Master Table

Headline list prices, mid-2026. ∞ = unlimited; £/GB blank where unlimited. Two tiers interpolated (flagged model.).

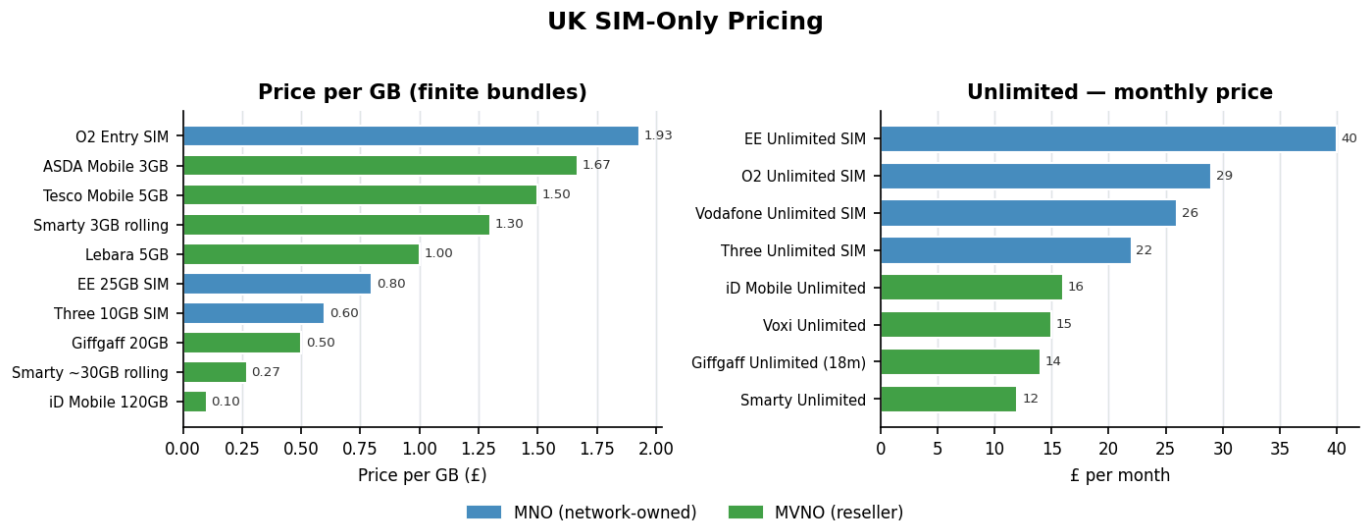
Operator	Type	Runs on	Plan	Data(GB)	£/mo	£/GB	Months	Basis
Three	MNO	Three	Unlimited SIM	∞	22.00	—	24	meas.
Three	MNO	Three	10GB SIM	10	6.00	0.60	1	meas.
EE	MNO	EE	Unlimited SIM	∞	40.00	—	24	meas.
EE	MNO	EE	25GB SIM	25	20.00	0.80	24	model.
O2	MNO	O2	Entry SIM	3	5.80	1.93	1	meas.
O2	MNO	O2	Unlimited SIM	∞	29.00	—	24	model.
Vodafone	MNO	Vodafone	Unlimited SIM	∞	26.00	—	24	meas.
Smarty	MVNO	Three	3GB rolling	3	3.90	1.30	1	meas.
Smarty	MVNO	Three	~30GB rolling	30	8.00	0.27	1	meas.
Smarty	MVNO	Three	Unlimited	∞	12.00	—	12	meas.
iD Mobile	MVNO	Three	120GB	120	12.00	0.10	1	meas.
iD Mobile	MVNO	Three	Unlimited	∞	16.00	—	1	meas.
Giffgaff	MVNO	O2	20GB	20	10.00	0.50	1	meas.
Giffgaff	MVNO	O2	Unlimited (18m)	∞	14.00	—	18	meas.
Lebara	MVNO	Vodafone	5GB	5	5.00	1.00	1	meas.
Voxi	MVNO	Vodafone	Unlimited	∞	15.00	—	1	meas.
Tesco Mobile	MVNO	O2	5GB	5	7.50	1.50	12	meas.
ASDA Mobile	MVNO	Vodafone	3GB	3	5.00	1.67	1	meas.

Read: cheapest effective data is an MVNO large bundle (iD 120GB ≈ £0.10/GB); network-owned own-brand plans carry a 3–15× per-GB premium.

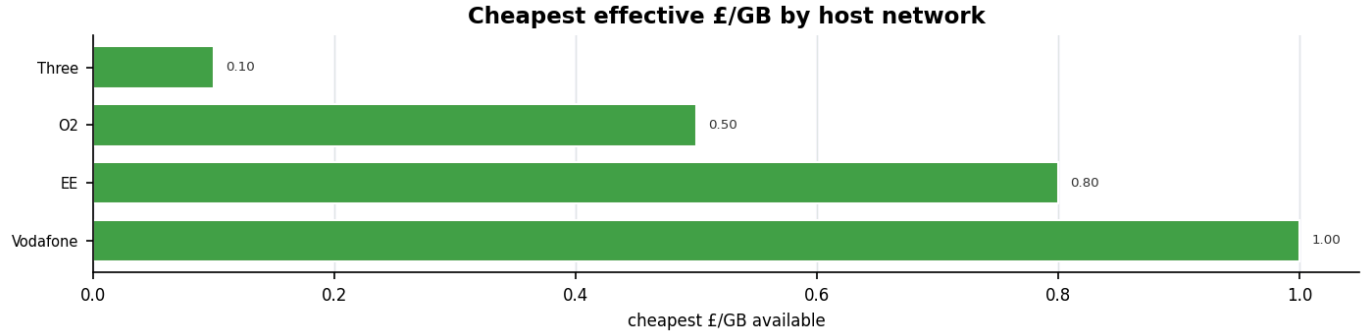
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Pricing Analysis & Charts

Price per GB (finite bundles) and unlimited monthly price, coloured MNO vs MVNO.



Cheapest effective £/GB available on each physical network — the Three network anchors the market floor.



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Price-per-GB League Table

Every finite-allowance tariff, ranked cheapest effective data first. Unlimited plans excluded (per-GB undefined).

#	Operator	Plan	GB	£/mo	£/GB	Type
1	iD Mobile	120GB	120	12.00	0.10	MVNO
2	Smarty	~30GB rolling	30	8.00	0.27	MVNO
3	Giffgaff	20GB	20	10.00	0.50	MVNO
4	Three	10GB SIM	10	6.00	0.60	MNO
5	EE	25GB SIM	25	20.00	0.80	MNO
6	Lebara	5GB	5	5.00	1.00	MVNO
7	Smarty	3GB rolling	3	3.90	1.30	MVNO
8	Tesco Mobile	5GB	5	7.50	1.50	MVNO
9	ASDA Mobile	3GB	3	5.00	1.67	MVNO
10	O2	Entry SIM	3	5.80	1.93	MNO

The top of the table is dominated by large-bundle MVNOs on Three; own-brand MNO small bundles sit at the bottom, paying a premium per gigabyte for brand and support.

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Best Value by User Profile

Which tariff wins for a representative light / medium / heavy data user, on measured list prices.

User profile	Typical need	Best-value pick (this dataset)	£/mo	Why
Light (≤3GB)	calls/texts + light browsing	Smarty 3GB	3.90	cheapest small bundle on a premium network
Medium (~20-30GB)	daily streaming on mobile	Smarty ~30GB	8.00	≈£0.27/GB, no in-contract rises
Heavy (100GB+)	hotspot / primary connection	iD Mobile 120GB	12.00	≈£0.10/GB — cheapest effective data here
Unlimited	tethering / no metering	Smarty Unlimited	12.00	lowest unlimited in the set
Premium/coverage	max coverage + perks	EE / Vodafone unlimited	26-40	brand, roaming, support certainty

These picks are from the tariffs in this report, not the whole market. Send a target user mix and I can compute a weighted best-value basket.

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Operator Profile — EE

The premium-coverage network · MNO · network: own (BT Group)

EE is the consumer mobile brand of BT Group and consistently tops UK coverage and speed rankings. It positions at the premium end: its unlimited SIM sits highest among the MNOs at ~£40/month.

EE rarely competes on headline price; instead it bundles perks (data gifting, roaming, device insurance) and leans on network quality. Value-seeking customers are steered to no MVNO of its own on the same masts, so the price floor on EE's network is EE itself.

Tariffs in this report

Plan	Data(GB)	£/mo	£/GB	Contract
Unlimited SIM	∞	40.00	—	24m
25GB SIM	25	20.00	0.80	24m

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Operator Profile — O2 (VMO2)

The wholesale host · MNO · network: own (Virgin Media O2)

O2 is part of Virgin Media O2, which closed 2024 with 45.7M total connections. Its own-brand entry SIM starts around £5.80 and EU roaming runs across the range.

O2's strategic weight is as a wholesale host: Giffgaff, Tesco Mobile and Sky Mobile all ride O2's network. Sky Mobile + Tesco Mobile alone hold >15% of the consumer segment, so O2 monetises both retail and wholesale layers.

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Operator Profile — Vodafone

The converging incumbent · MNO · network: own

Vodafone offers an unlimited SIM around £26/month and hosts a broad value layer — Voxi (youth, social-data-free), Lebara (international calling) and ASDA Mobile.

Vodafone's UK trajectory is dominated by its consolidation with Three; the combined network reshapes coverage and capacity, and is the single biggest structural change in the market over the report horizon.

Tariffs in this report

Plan	Data(GB)	£/mo	£/GB	Contract
Unlimited SIM	∞	26.00	—	24m

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Operator Profile — Three

The price leader · MNO · network: own

Three is the aggressive price leader among the MNOs: unlimited at ~£22/month undercuts every rival, and its network hosts the cheapest per-GB resellers in the country.

Smarty and iD Mobile both run on Three; iD Mobile's 120GB at £12 (~£0.10/GB) is the cheapest effective data in this report. The Three network therefore anchors the market's price floor.

Tariffs in this report

Plan	Data(GB)	£/mo	£/GB	Contract
Unlimited SIM	∞	22.00	—	24m
10GB SIM	10	6.00	0.60	1m

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MVNO Profile — Smarty

Reseller · runs on Three

Data-focused, no-frills, no in-contract price rises. 3GB from £3.90, ~30GB at £8 (~£0.27/GB), unlimited £12. One of the fastest-growing ad spenders (+90% YoY 2023), reflecting an aggressive challenger push on Three's network.

Tariffs in this report

Plan	Data(GB)	£/mo	£/GB	Contract
3GB rolling	3	3.90	1.30	1m
~30GB rolling	30	8.00	0.27	1m
Unlimited	∞	12.00	—	12m

> Because MVNOs buy wholesale capacity from the host MNO, their pricing floor is set by that wholesale rate — which is why the cheapest per-GB deals cluster on the network with the most aggressive wholesale terms (Three).

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MVNO Profile — iD Mobile

Reseller · runs on Three

Owned by Currys/Dixons. Data-rollover and the market's cheapest large bundle — 120GB for £12 (~£0.10/GB); unlimited £16. Aggressive on high-data tiers, and the single cheapest effective £/GB in this report.

Tariffs in this report

Plan	Data(GB)	£/mo	£/GB	Contract
120GB	120	12.00	0.10	1m
Unlimited	∞	16.00	—	1m

> Because MVNOs buy wholesale capacity from the host MNO, their pricing floor is set by that wholesale rate — which is why the cheapest per-GB deals cluster on the network with the most aggressive wholesale terms (Three).

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MVNO Profile — Giffgaff

Reseller · runs on O2

Community-run on O2's 99%-coverage network. 20GB £10, unlimited (18-month) £14, plus an 'Always On' SIM that throttles to 384kbps after 80GB. Strong brand loyalty and a distinctive member-referral acquisition model that keeps CPA low.

Tariffs in this report

Plan	Data(GB)	£/mo	£/GB	Contract
20GB	20	10.00	0.50	1m
Unlimited (18m)	∞	14.00	—	18m

> Because MVNOs buy wholesale capacity from the host MNO, their pricing floor is set by that wholesale rate — which is why the cheapest per-GB deals cluster on the network with the most aggressive wholesale terms (Three).

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MVNO Profile — Tesco Mobile

Reseller · runs on O2

Clubcard-linked loyalty, family perks. 5GB at £7.50; one of the largest MVNOs and, with Sky Mobile, part of the >15% consumer share on O2. Retail footprint gives it a low-cost acquisition channel.

Tariffs in this report

Plan	Data(GB)	£/mo	£/GB	Contract
5GB	5	7.50	1.50	12m

> Because MVNOs buy wholesale capacity from the host MNO, their pricing floor is set by that wholesale rate — which is why the cheapest per-GB deals cluster on the network with the most aggressive wholesale terms (Three).

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MVNO Profile — Voxi

Reseller · runs on Vodafone

Vodafone's youth brand with 'endless' social-media data on some plans. Unlimited data £15; SIM-only, 30-day rolling, no credit check. Designed to defend the under-30 segment against challenger MVNOs.

Tariffs in this report

Plan	Data(GB)	£/mo	£/GB	Contract
Unlimited	∞	15.00	—	1m

> Because MVNOs buy wholesale capacity from the host MNO, their pricing floor is set by that wholesale rate — which is why the cheapest per-GB deals cluster on the network with the most aggressive wholesale terms (Three).

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MVNO Profile — Lebara

Reseller · runs on Vodafone

International-calling specialist (100+ minutes to 42 countries on entry plans). 5GB £5; frequent promotional unlimited at ~£10 for new customers. Targets diaspora and value segments.

Tariffs in this report

Plan	Data(GB)	£/mo	£/GB	Contract
5GB	5	5.00	1.00	1m

> Because MVNOs buy wholesale capacity from the host MNO, their pricing floor is set by that wholesale rate — which is why the cheapest per-GB deals cluster on the network with the most aggressive wholesale terms (Three).

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MVNO Profile — ASDA Mobile

Reseller · runs on Vodafone

Supermarket value brand on Vodafone masts. 3GB £5; positioned on simplicity and low headline entry price rather than large bundles. Distribution via ASDA stores lowers acquisition friction.

Tariffs in this report

Plan	Data(GB)	£/mo	£/GB	Contract
3GB	3	5.00	1.67	1m

> Because MVNOs buy wholesale capacity from the host MNO, their pricing floor is set by that wholesale rate — which is why the cheapest per-GB deals cluster on the network with the most aggressive wholesale terms (Three).

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MVNO Profile — Sky Mobile

Reseller · runs on O2

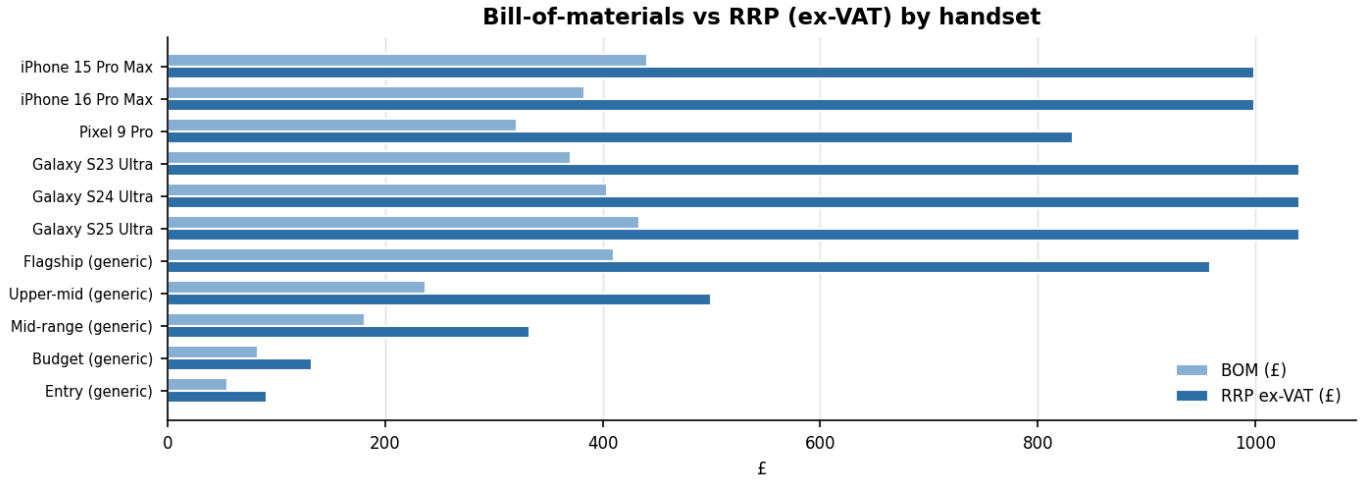
Bundled with Sky TV/broadband; data-rollover 'Piggybank'. Part of the >15% consumer segment on O2; monetises the wider Sky subscription base and cross-sells against churn.

> Because MVNOs buy wholesale capacity from the host MNO, their pricing floor is set by that wholesale rate — which is why the cheapest per-GB deals cluster on the network with the most aggressive wholesale terms (Three).

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Handset Market Overview

Handset economics matter to an operator because the device subsidy and trade-in flows sit alongside the airtime margin. This report models the cost-to-retail waterfall for a spread of flagships (Apple, Samsung, Google) and generic tier midpoints.



Bill-of-materials is the measured anchor (teardown estimates, analyst-dependent). Everything downstream — ex-factory, margin, markup — is derived from the stated model, so the whole column re-prices from a single constant.

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Handset Ex-Factory Margins — Master Table

BOM measured (teardown est.); ex-factory **modelled** = BOM×1.30. RRP shown inc- and ex-VAT. FX £0.79/USD, VAT 20%.

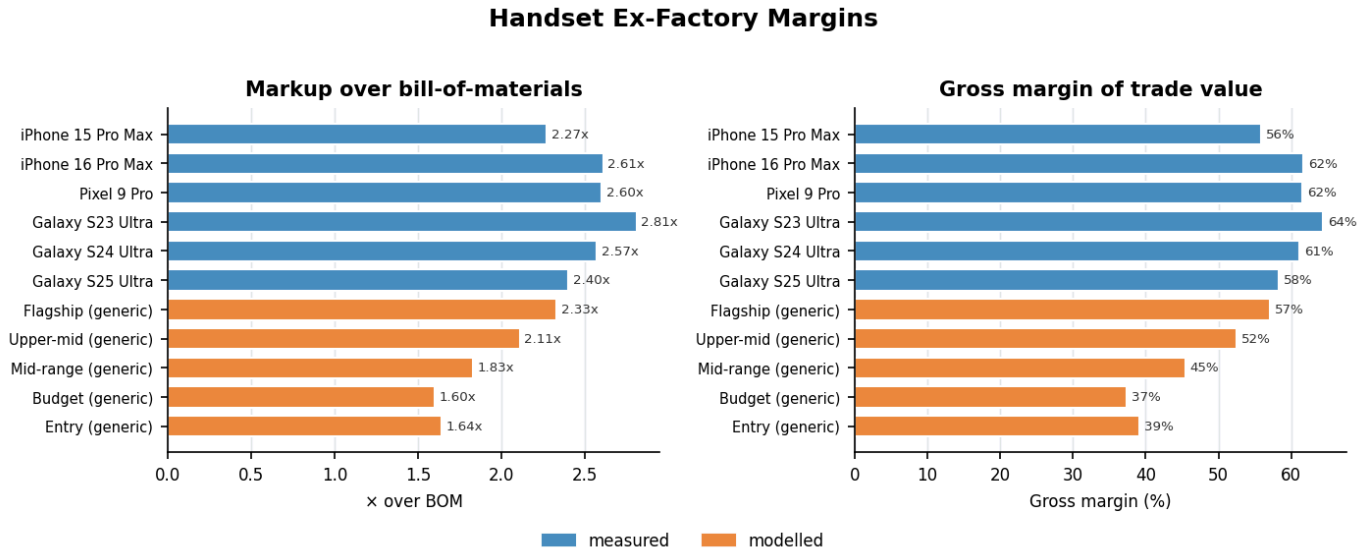
Model	Brand	Yr	BOM$	BOM£	Ex-fac£*	RRP£inc	RRP£exVAT	Gross mgn	Markup	Basis
iPhone 15 Pro Max	Apple	2023	558	440.82	573.07	1199	999.17	55.9%	2.27×	meas.
iPhone 16 Pro Max	Apple	2024	485	383.15	498.10	1199	999.17	61.7%	2.61×	meas.
Pixel 9 Pro	Google	2024	406	320.74	416.96	999	832.50	61.5%	2.60×	meas.
Galaxy S23 Ultra	Samsung	2023	469	370.51	481.66	1249	1040.83	64.4%	2.81×	meas.
Galaxy S24 Ultra	Samsung	2024	512	404.48	525.82	1249	1040.83	61.1%	2.57×	meas.
Galaxy S25 Ultra	Samsung	2025	549	433.71	563.82	1249	1040.83	58.3%	2.40×	meas.
Flagship (generic)	-	2026	520	410.80	534.04	1150	958.33	57.1%	2.33×	model.
Upper-mid (generic)	-	2026	300	237.00	308.10	599	499.17	52.5%	2.11×	model.
Mid-range (generic)	-	2026	230	181.70	236.21	399	332.50	45.4%	1.83×	model.
Budget (generic)	-	2026	105	82.95	107.84	159	132.50	37.4%	1.60×	model.
Entry (generic)	-	2026	70	55.30	71.89	109	90.83	39.1%	1.64×	model.

***Ex-factory is modelled, not a quoted number.** BOM excludes assembly, software, IP/royalties, logistics, warranty and marketing — gross margin is the pool those are paid from, not net profit.

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Handset Margin Analysis

Markup over bill-of-materials and gross margin of trade value, coloured measured vs modelled.



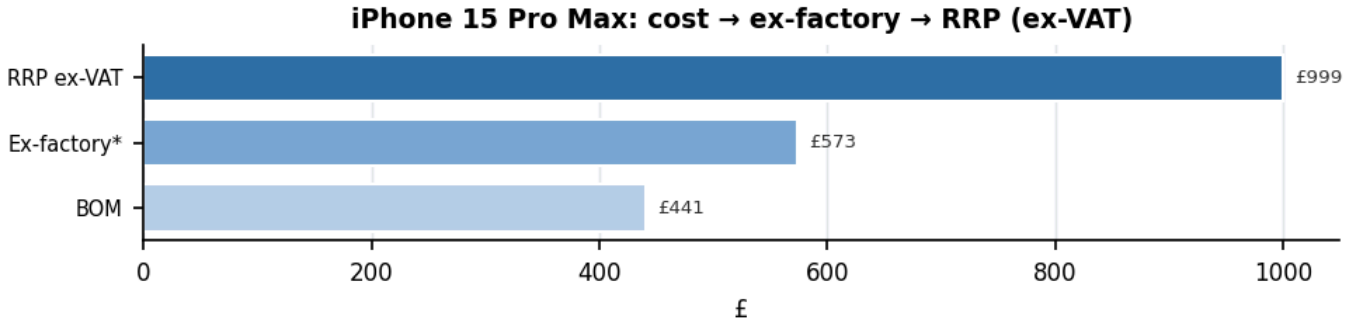
- Flagships mark up ~2.3–2.8× their component cost; the newest Galaxy Ultra tops the set.
- Margin compresses toward budget tiers (~1.6×) — cheaper phones leave less overhead headroom.
- Apple and Samsung flagships carry similar BOM (~£370–440); RRP dispersion drives the margin gap.
- Google Pixel 9 Pro undercuts on BOM (~£320) at a lower RRP — a different margin posture.

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Handset — iPhone 15 Pro Max

Apple · 2023 · basis: measured (teardown est. (analyst-dependent))

Apple's 2023 titanium flagship on the A17 Pro. Teardown estimates put the BOM near the top of the set; display and camera systems are the costliest single components. Sold at a premium RRP, it anchors the high end of the margin curve.



BOM $	BOM £	Ex-factory £*	RRP £ inc-VAT	RRP £ ex-VAT	Gross margin	Markup ×BOM
558	440.82	573.07	1199	999.17	55.9%	2.27×

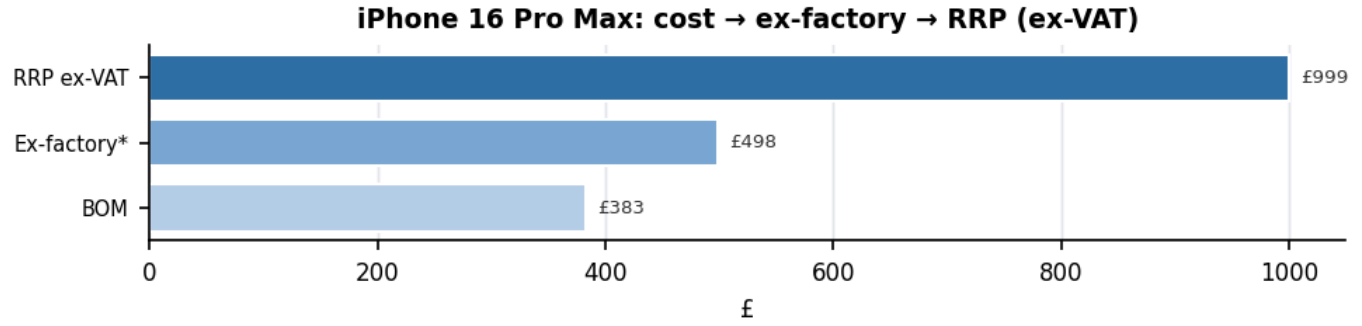
*Ex-factory modelled at BOM×1.30. BOM is the measured anchor; margin and markup are derived. RRP is the public UK launch price.

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Handset — iPhone 16 Pro Max

Apple · 2024 · basis: measured (Counterpoint BoM ~$485)

The 2024 A18 Pro flagship. Counterpoint estimated a BOM around $485, with display and rear-camera each ~16% of components. Held at the same RRP as its predecessor, so the margin depends heavily on the analyst's BOM figure.



BOM $	BOM £	Ex-factory £*	RRP £ inc-VAT	RRP £ ex-VAT	Gross margin	Markup ×BOM
485	383.15	498.10	1199	999.17	61.7%	2.61×

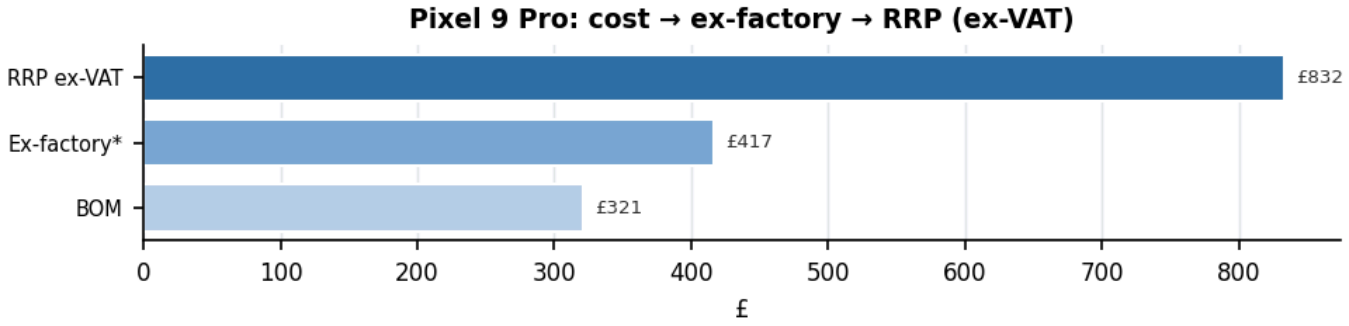
*Ex-factory modelled at BOM×1.30. BOM is the measured anchor; margin and markup are derived. RRP is the public UK launch price.

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Handset — Pixel 9 Pro

Google · 2024 · basis: measured (BoM ~$400-406 (Tensor G4 $80))

Google's 2024 flagship on the in-house Tensor G4 (~$80) with a Samsung OLED (~$75). BOM (~$400-406) sits below Apple/Samsung Ultra flagships, and at a lower RRP it runs a deliberately different margin posture aimed at share and AI features.



BOM $	BOM £	Ex-factory £*	RRP £ inc-VAT	RRP £ ex-VAT	Gross margin	Markup ×BOM
406	320.74	416.96	999	832.50	61.5%	2.60×

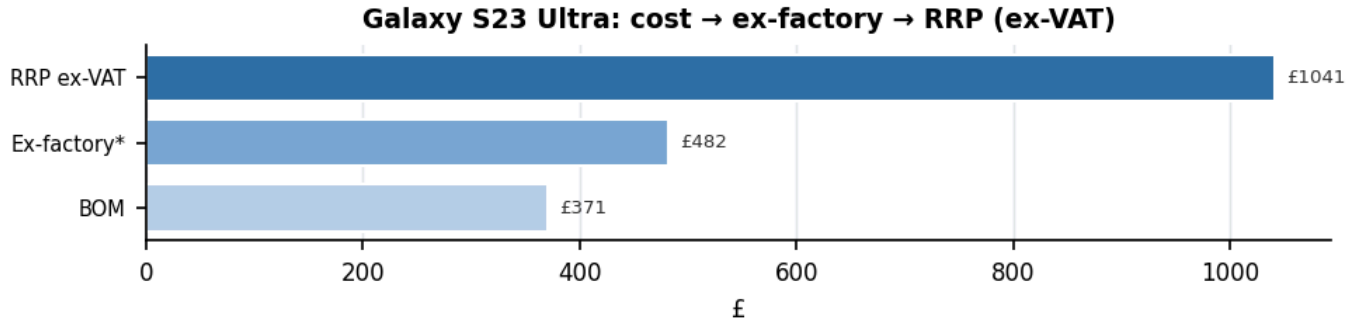
*Ex-factory modelled at BOM×1.30. BOM is the measured anchor; margin and markup are derived. RRP is the public UK launch price.

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Handset — Galaxy S23 Ultra

Samsung · 2023 · basis: measured (teardown est. $469)

Samsung's 2023 Ultra (Snapdragon 8 Gen 2). Widely-cited BOM ~$469; the 200MP camera and display drive component cost. Highest markup in this set at its launch RRP.



BOM $	BOM £	Ex-factory £*	RRP £ inc-VAT	RRP £ ex-VAT	Gross margin	Markup ×BOM
469	370.51	481.66	1249	1040.83	64.4%	2.81×

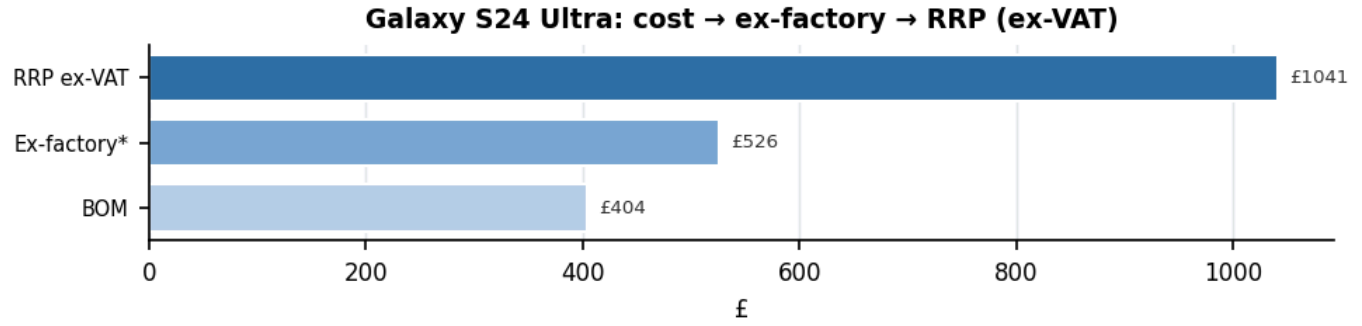
*Ex-factory modelled at BOM×1.30. BOM is the measured anchor; margin and markup are derived. RRP is the public UK launch price.

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Handset — Galaxy S24 Ultra

Samsung · 2024 · basis: measured (reported ~$512)

The 2024 Ultra (Snapdragon 8 Gen 3) added on-device AI; reported BOM ~$512, a modest rise on the S23 Ultra at an unchanged RRP, compressing margin slightly.



BOM $	BOM £	Ex-factory £*	RRP £ inc-VAT	RRP £ ex-VAT	Gross margin	Markup ×BOM
512	404.48	525.82	1249	1040.83	61.1%	2.57×

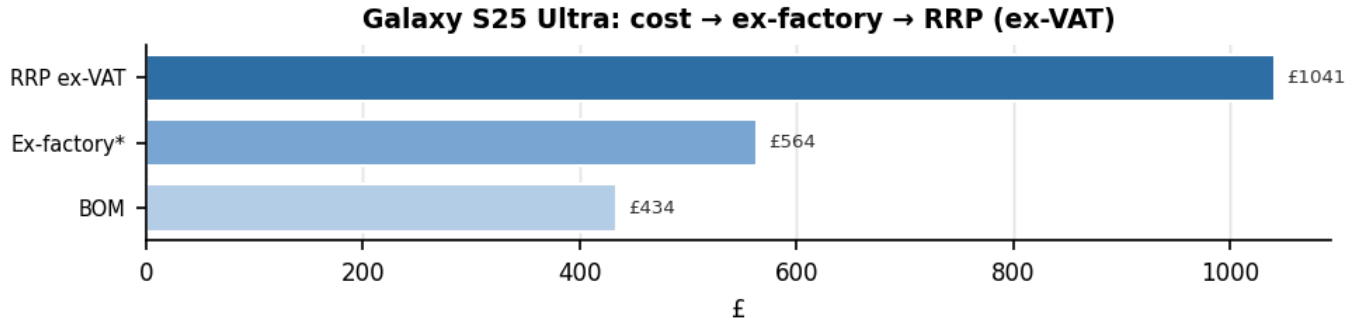
*Ex-factory modelled at BOM×1.30. BOM is the measured anchor; margin and markup are derived. RRP is the public UK launch price.

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Handset — Galaxy S25 Ultra

Samsung · 2025 · basis: measured (reported ~$549 (Snapdragon 8 Elite))

The 2025 Ultra moved to the Snapdragon 8 Elite, pushing reported BOM to ~$549 — the most expensive to build here. At an unchanged RRP the extra silicon cost is absorbed from margin.



BOM $	BOM £	Ex-factory £*	RRP £ inc-VAT	RRP £ ex-VAT	Gross margin	Markup ×BOM
549	433.71	563.82	1249	1040.83	58.3%	2.40×

*Ex-factory modelled at BOM×1.30. BOM is the measured anchor; margin and markup are derived. RRP is the public UK launch price.

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Model Sensitivity — Ex-Factory Multiplier

The one modelled lever in the handset waterfall is EXFACTORY_MULT (default 1.30). This table shows the estimated ex-factory price (£) for the measured handsets at three multipliers, so the assumption is transparent and tunable.

Model	× 1.15 (lean)	× 1.30 (base)	× 1.50 (rich)
iPhone 15 Pro Max	507	573	661
iPhone 16 Pro Max	441	498	575
Pixel 9 Pro	369	417	481
Galaxy S23 Ultra	426	482	556
Galaxy S24 Ultra	465	526	607
Galaxy S25 Ultra	499	564	651

Nothing else in the report depends on a hidden constant. Change EXFACTORY_MULT in build_report.py / gen_sample.py and the whole ex-factory column, and any downstream chart, re-price consistently.

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Marketing & Acquisition Economics

Customer acquisition is the swing factor in a low-ARPU market. UK carriers spend an estimated **15–20% of revenue** on acquisition and retention. Digital ad spend was ~$420M in 2023; challenger brands spent hardest (Smarty +90%, Three +60% YoY), reflecting an MVNO land-grab.

Structural shifts are cutting unit costs: eSIM removes plastic-SIM logistics and is estimated to cut acquisition cost ~30%, while online channels now drive ~35% of acquisition and grow ~18%/year. The one figure still modelled here is a headline per-SIM CPA (£18) — send your real number and it flows through the model.

Metric	Value	Unit	Period	Segment	Basis
Postpaid ARPU	15.73	GBP/month	2023	postpaid	meas.
Prepaid ARPU	5.18	GBP/month	2023	prepaid	meas.
VMO2 total connections	45.7	million	2024	all	meas.
5G SA population coverage	83	%	2025	all	meas.
Mobile data usage growth	18	% YoY	2025	all	meas.
MVNO consumer share (Sky+Tesco)	15	% (>)	2025	MVNO	meas.
Acquisition+retention spend	17.5	% of revenue	2025	all	meas.
UK telecom digital ad spend	420	USD million	2023	all	meas.
Smarty ad-spend growth	90	% YoY	2023	MVNO	meas.
Three ad-spend growth	60	% YoY	2023	MNO	meas.
eSIM acquisition-cost saving	30	% (~)	2025	all	meas.
Online share of acquisition	35	%	2025	all	meas.
Cost per acquisition (SIM)	18	GBP (est)	2026	all	model.

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Segment & Structural Context

Two structural forces dominate the horizon. First, the **Vodafone–Three consolidation** reshapes network capacity and the competitive floor: Three has been the price leader and hosts the cheapest resellers, so the merged entity's tariff posture matters to the whole market.

Second, the **MVNO layer keeps taking consumer share** — Sky Mobile + Tesco Mobile alone exceed 15% — pressuring own-brand ARPU while giving the host MNOs (especially O2) a wholesale revenue line. The main-brand subscriber base of several MNOs is forecast to keep shrinking as value brands absorb price-sensitive users.

For an operator building on this market, the implication is clear: differentiate on quality, bundle perks and wholesale, because the per-GB race to the bottom is already won by large-bundle MVNOs.

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Data Dictionary

plans

Column	Meaning
operator/type/host	brand, MNO|MVNO, physical network
plan / data_gb	tariff label; GB or 'unlimited'
price_gbp_month	headline £/month
price_per_gb_gbp	price ÷ data_gb; blank if unlimited
contract_months	1 = 30-day rolling
basis	measured | modelled

handsets

Column	Meaning
bom_usd/gbp	bill-of-materials (teardown est.)
est_exfactory_gbp	modelled = BOM × 1.30
rrp_ex_vat_gbp	RRP ÷ 1.20 (trade value)
gross_margin_pct	(rrp_exvat − bom)/rrp_exvat
markup_x_over_bom	rrp_exvat ÷ bom
basis	measured | modelled

market

Column	Meaning
metric/value/unit	the indicator and its figure
period/segment	year; postpaid|prepaid|MVNO|all
basis/source	measured|modelled; provenance

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The Report-Engine Sample Feed

The whole report is generated from the datasets by build_report.py; the tabular sample is emitted by gen_sample.py (JSON checksum a792719cdd4e03a6).

File	What it is
plans.csv	SIM-only tariffs + £/GB
handsets.csv	handset cost→ex-factory→RRP waterfalls
market.csv	market / marketing metrics
uk_data_market_sample.json	all three tables + meta & assumptions
gen_sample.py	emits CSVs+JSON; --check verifies
build_report.py	emits this long report; scales with the data
gen_charts.py / charts	the figures used throughout
data_dictionary.md	column schema + assumptions

> To grow this to 60–100 pages: add operators/handsets/tiers to the datasets (each new operator ≈ 1 page, each MVNO pair ≈ 1 page), or send your report engine's own dataset and template and I'll drive it 1:1.

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Glossary

Term	Definition
MNO	Mobile Network Operator — owns the physical radio network (EE, O2, Vodafone, Three).
MVNO	Mobile Virtual Network Operator — resells airtime on an MNO's network.
ARPU	Average Revenue Per User — monthly revenue ÷ subscribers.
BOM	Bill of Materials — component cost of a device (excludes assembly, software, overhead).
Ex-factory	Price a device leaves the factory at; modelled here as BOM × 1.30.
RRP	Recommended Retail Price; shown inc- and ex-VAT.
5G SA	5G Standalone — 5G core, not anchored to 4G.
CPA	Cost Per Acquisition — marketing spend to win one customer.
eSIM	Embedded SIM — provisioned over the air, no plastic SIM.
£/GB	Effective price per gigabyte = monthly price ÷ data allowance.

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Sources & Method

- SIM-only pricing — UK operator price lists & comparison sites, mid-2026 (measured).
- Handset BOM — Counterpoint / TechInsights / IHS-style teardown estimates, 2023–2025 (measured; analyst-dependent).
- ARPU — Ofcom / Statista, 2023 (measured).
- 5G coverage & data growth — Ofcom Connected Nations 2025 (measured).
- MVNO share & VMO2 connections — market reports / Virgin Media O2, 2024–25 (measured).
- Marketing & acquisition — SensorTower / IBISWorld / industry estimates, 2023–25 (measured ranges).
- Cost-per-acquisition (£) — modelled placeholder, awaiting client figure.

Modelling constants: FX £0.79/USD · ex-factory ×1.30 · VAT 20% — all editable in the generator; change one and the dataset re-prices.

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Appendix A — Raw Tariff Table

Operator	Type	Host	Plan	GB	£/mo	Contract(m)
Three	MNO	Three	Unlimited SIM	∞	22.00	24
Three	MNO	Three	10GB SIM	10	6.00	1
EE	MNO	EE	Unlimited SIM	∞	40.00	24
EE	MNO	EE	25GB SIM	25	20.00	24
O2	MNO	O2	Entry SIM	3	5.80	1
O2	MNO	O2	Unlimited SIM	∞	29.00	24
Vodafone	MNO	Vodafone	Unlimited SIM	∞	26.00	24
Smarty	MVNO	Three	3GB rolling	3	3.90	1
Smarty	MVNO	Three	~30GB rolling	30	8.00	1
Smarty	MVNO	Three	Unlimited	∞	12.00	12
iD Mobile	MVNO	Three	120GB	120	12.00	1
iD Mobile	MVNO	Three	Unlimited	∞	16.00	1
Giffgaff	MVNO	O2	20GB	20	10.00	1
Giffgaff	MVNO	O2	Unlimited (18m)	∞	14.00	18
Lebara	MVNO	Vodafone	5GB	5	5.00	1
Voxi	MVNO	Vodafone	Unlimited	∞	15.00	1
Tesco Mobile	MVNO	O2	5GB	5	7.50	12
ASDA Mobile	MVNO	Vodafone	3GB	3	5.00	1

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Appendix B — Handset Reference

Model	Brand	Year	BOM$	RRP£inc	Note / source
iPhone 15 Pro Max	Apple	2023	558	1199	teardown est. (analyst-dependent)
iPhone 16 Pro Max	Apple	2024	485	1199	Counterpoint BoM ~$485
Pixel 9 Pro	Google	2024	406	999	BoM ~$400-406 (Tensor G4 $80)
Galaxy S23 Ultra	Samsung	2023	469	1249	teardown est. $469
Galaxy S24 Ultra	Samsung	2024	512	1249	reported ~$512
Galaxy S25 Ultra	Samsung	2025	549	1249	reported ~$549 (Snapdragon 8 Elite)
Flagship (generic)	-	2026	520	1150	tier midpoint
Upper-mid (generic)	-	2026	300	599	tier midpoint
Mid-range (generic)	-	2026	230	399	tier midpoint
Budget (generic)	-	2026	105	159	tier midpoint
Entry (generic)	-	2026	70	109	tier midpoint

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